

Our Carbon Credit Ratings Framework for **Biochar Projects**

Incentivizing investment in real climate action

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Introduction

Sylvera Ratings are the most reliable and trustworthy in the market.

Sylvera has developed a rigorous bottom-up approach in order to produce the most accurate ratings and analyses for projects in the carbon markets.

What sets Sylvera apart

- **Project-type-specific frameworks:** We build rigorous frameworks and production systems for every project category to accurately test project design, carbon accounting, and climate impact claims.

Sylvera's frameworks are peer-reviewed by a committee of experts and carbon market stakeholders – including project developers & registries – to ensure scientific consensus. We have published this document so buyers understand how we developed our unique, nuanced framework, and we have included some examples of exactly what we test and how we do it.

[Read our white paper for more information.](#)

- **Unparalleled depth & accuracy:** We extract, clean, and organize data from project design documentation (PDD) and every monitoring report. Then we meticulously build carbon, strength of baseline and financial additionality models from the ground up to validate emissions reductions or removals claims and evaluate project economics.

Our project assessments are the most comprehensive in the market, providing granular analysis of core project characteristics, insightful data visualizations, and interactive maps.

- **Independent Data Validation:** Our expert analysts leverage advanced machine learning (ML) technology, verified, independent data, and proprietary field data to test the accuracy of credit issuances and claims.

The comparison of independent data specific to each project against the data reported in the project's documentation is the cornerstone of high quality due diligence.

Key terms and concepts

Biochar	Biochar is a stable organic carbon compound formed through pyrolysis.
Biochar Decay Rate	Decay rate refers to the speed at which biochar's stored carbon is released back into the atmosphere due to environmental and external factors such as soil conditions and temperature, application methods, and natural risks.
Biochar Stability	Refers to the inherent ability of biochar to retain sequestered carbon over time, based on its intrinsic characteristics and resistance to chemical, biological, and physical degradation.
Carbon credit	A tradable unit representing one metric ton of carbon dioxide (CO ₂), or an equivalent amount of another greenhouse gas (GHG), avoided or removed from Earth's atmosphere.
Gasification	A thermochemical conversion process that heats biomass at high temperatures (800–1,000°C) with a controlled amount of oxygen or steam, producing syngas and a smaller quantity of biochar than pyrolysis.
Greenhouse Gas (GHG) Emissions	Refers to the release of gases into the atmosphere that trap heat and contribute to climate change.
Hydrogen to Carbon Ratio (H/Corg)	The hydrogen (H) to organic carbon (Corg) molar ratio correlates with the stability of biochar.
Internal Rate of Return (IRR)	Internal Rate of Return used for investment analysis to estimate the return on an investment.
Life Cycle Assessment (LCA)	This refers to the analysis conducted to estimate the amount of emissions associated with the sourcing of biomass, the creation of biochar and its deployment.
Over-crediting risk	This refers to the risk that the project has issued credits in excess of what is justifiable against the business as usual scenario.
Organic Carbon (Corg)	The organic fraction of total carbon in biochar, measured as a percentage of dry weight. It is calculated as total carbon minus inorganic carbon content.
Permanence Adjustment Factor	This refers to the percentage of carbon that will stay secure in the biochar for the next 200+ years.
Project emissions	Emissions associated with ongoing operations of the carbon credit project.
Pyrolysis	Pyrolysis is the heating/burning of an element in an oxygen-absent, controlled environment.
Random Reflectance	A measure of the optical reflectivity of carbon structures under microscopic analysis, derived from coal petrology. Higher values indicate greater thermal maturity and aromaticity, which are associated with increased long-term carbon stability.
Vintage	This refers to the year, or timeframe, associated with an issued carbon credit.

The state of Biochar

The challenge

Biochar is an essential technology to help scale the engineered Carbon Dioxide Removal (CDR) market and meet the growing demand for carbon removal credits. The majority of delivered CDR credits, ones which are validated and issued, are from biochar projects. This means that biochar credits are some of the only technology CDR credits companies can purchase and retire to compensate for their emissions.

Biochar developers and registries are nascent and ultimately have yet to face the same scrutiny as other sectors of the market. For biochar projects there is a notable deficiency of data available around key project performance parameters. The rush to procure biochar credits is understandable, but without transparency and deep due diligence the credits represent a material risk in your carbon credit portfolio.

With the right data disclosures and collaboration with developers, we can provide the deep project-level insights needed to inform buyers' investment due diligence and help bring **these technologies** to scale faster. If you're a project developer interested in having Sylvera rate your project to help prove impact and optimize investment, reach out to our [team](#).

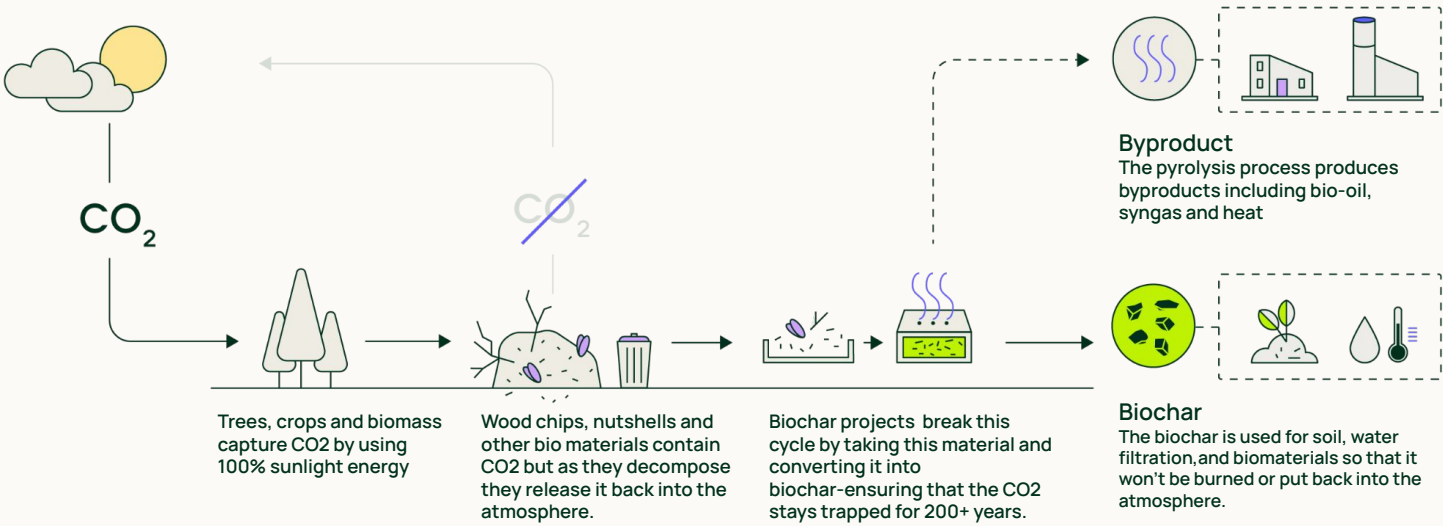
The solution

To overcome the challenge of data transparency and scale the biochar market, the following data points are needed to ensure trust in biochar credits:

1. **Investment Analysis Parameters:** Disclosing financial parameters allows the market to assess whether carbon finance is necessary for the project to be economic. Without this data, the financial additionality is uncertain.
2. **Biochar Carbon Content:** The verifiable amount of organic carbon present in the biochar created is the foundation of the net carbon sequestration calculations.
3. **H/Corg Ratio or Random Reflectance:** Lab-based indicators used to assess the degree of carbonization and the stability of biochar carbon and, where relevant under the methodology, to support credit quantification.
4. **H/C Ratio:** The ratio of hydrogen to organic carbon, based on lab analysis, is a key factor in the quantification of the volume of credits issued and the stability of carbon sequestered.
5. **Feedstock:** Disclosing the feedstock type, sourcing principles and feedstock counterfactual related to the project is important in understanding impacts associated with feedstocks, and this can have a big impact on the net removal calculation.
6. **LCA Emissions:** Life cycle assessments (LCA) demonstrate whether emissions associated with the creation and deployment of biochar have been accounted for in full. Emissions at each stage of the life cycle should be made available to determine any risk of over-crediting.
7. **End-use Traceability:** Project developers should maintain robust traceability of biochar end-use applications, with practices such as multi-layer verification and geographic coordinate disclosure to enable effective monitoring, risk assessment, and detection of potential reversals or inappropriate use.

What is a Biochar project?

Biochar projects produce **removals credits** through the thermochemical conversion of biomass, such as pyrolysis and gasification, in an oxygen-limited environment. Not all carbon is converted to biochar during the thermochemical process; co-products include bio-oil, pyrolysis gases, and heat. The biochar produced is abundant in carbon, stable and has numerous applications. Its stability is primarily determined by the production temperature and the resulting H/Corg ratio.



The input to the process is primarily plant-based and animal-derived biomass. The biomass can be sourced locally from the surrounding forests or agricultural lands and may come from many sources, for example a forest management company with wood chips or a farmer with grass and crop residue. The sourced biomass has a direct impact on the quality of the biochar created. This input typically requires pre-treatment, including drying, before entering the thermochemical conversion process.

The reactor heats the biomass in an oxygen-limited environment at temperatures typically ranging from 300°C to 800°C. The process creates other co-products such as bio-oil, syngas and excess thermal energy, which can be sold for additional revenue streams. Alternative revenue streams, in addition to those derived from the sale of carbon credits, have meaningful implications for the **additionality of biochar projects** as these revenue streams impact the extent to which carbon finance is required for these projects to be economic.

Biochar alternative use cases

Water filters & wastewater treatment	Building materials	Renewable heat
Added to soil (fertilizer, compost additive)	Animal farming (feed, slurry treatment)	Fabric additive in textiles

Biochar is predominantly applied to soil, where it improves soil fertility through enhanced nutrient retention, water holding capacity, and microbial activity. Innovative non-soil applications, such as incorporation into building materials, are emerging as alternative carbon storage pathways.

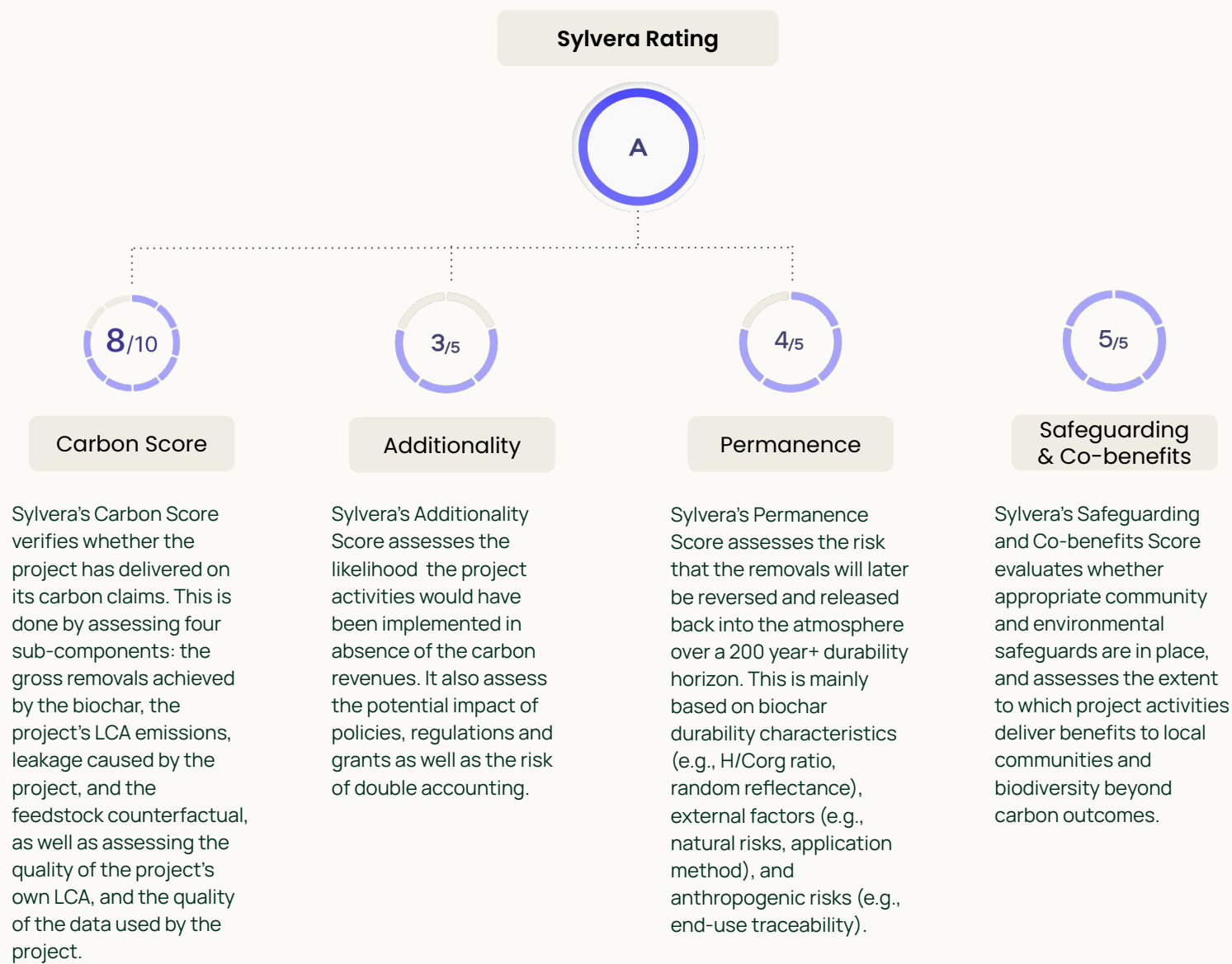
Current scientific consensus primarily determines biochar decay rates using H/Corg ratio and ambient soil temperature as key parameters. Emerging approaches such as random reflectance propose longer durability claims (1000+ years), though the science remains contested and unstandardized. Over-crediting risk arises when a project's reported removals exceed what consensus-based permanence models indicate for equivalent conditions, or when reported input data fails cross-validation against independently verified sources. Traceability from feedstock to end-use is also essential to verify that credited removals are real and permanent.

A reminder of our scoring pillars

Our top level Sylvera Ratings (**AAA, AA, A, BBB, BB, B, C, D**) reflect whether each credit associated with the project is likely to remove 1 metric ton of CO₂e emissions.

This rating is derived from a combination of scores that assess the **carbon** performance, **additionality** and **permanence** of the project. The scores in these three core pillars are combined in a series of matrices to ensure that underperformance in one key area does not get overshadowed by high performance in others.

Co-benefits are also assessed but they do not feed into the Sylvera Rating, as they do not have a direct bearing on the climate impact of carbon credits. Including them in the Sylvera Rating could lead to a high co-benefits score obscuring poor performance on carbon removal. Aspects of the project relating to co-benefits that could materially impact the project's ability to deliver it's stated climate benefit are, however, reflected in the Sylvera Rating.



Carbon Score

What is it?

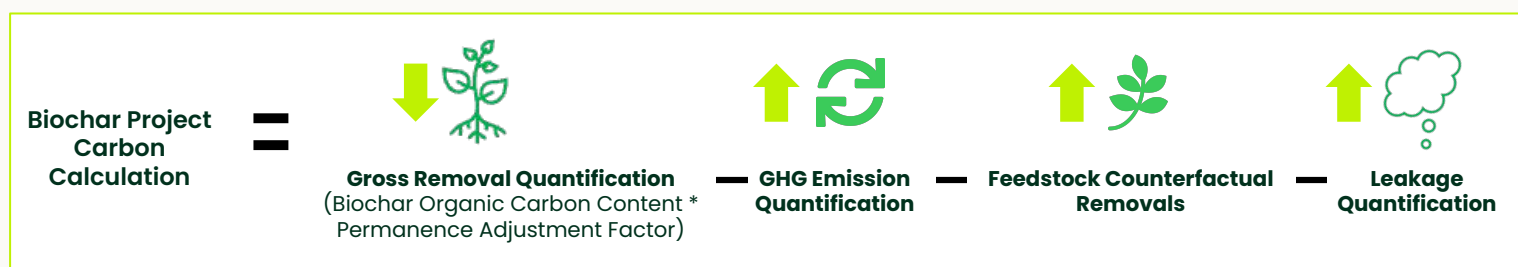
Sylvera's Carbon Score verifies whether a project is accurately reporting on the carbon removals achieved by the activity.

Note: The Carbon Score must be considered alongside the Additionality and Permanence Scores.

Why does it matter?

Accurate carbon accounting underpins the validity of a project's issuance and material under or over reporting of emissions will impact the number of credits that have been issued. If Sylvera-detected project carbon is significantly lower than the project's reported figures, reported carbon removal amounts are likely inaccurate and there is a higher risk of over-issuance.

How do we calculate the Carbon Score?



Gross Removal Quantification

Sylvera independently evaluates gross removals by reviewing existing decay models across peer-reviewed literature and carbon registries. While most follow [Woolf et al. \(2021\)](#), which uses H/Corg ratio and soil temperature as primary parameters, differences in regression methods and datasets produce varying estimates. Sylvera assesses the full range of decay model outputs over a 200 year+ durability horizon using H/Corg ratio as the baseline parameter, with independently modeled soil temperature data from the [Copernicus Climate Change Service \(C3S\) \(2022\)](#). Projects using advanced approaches, such as random reflectance for 1000+ year claims, trigger a case-by-case review by Sylvera.

$$\text{Gross Removal} = Q_{\text{biochar}} \times C_{\text{org}} \times F_{p, TH, Ts} \times 44/12$$

Where:

Q_{biochar} is the amount of biochar produced over the reporting period.

C_{org} is the organic carbon content of the biochar produced.

$F_{p, TH, Ts}$ is the permanence factor of biochar organic carbon over a given time horizon TH in a given soil at temperature TS .

$44/12$ is the value used to convert mass of carbon to mass of carbon dioxide emissions.

Sylvera cross-checks Q_{biochar} , C_{org} , and H/Corg ratio against validated project documentation to ensure data integrity. The assessment compares project-reported gross removals against the range of plausible estimates from available decay models under similar biochar characteristics and environmental conditions. Divergence may indicate over-crediting risk if a project issues significantly more credits than peer comparisons would suggest for equivalent inputs. Where reported quantities fall within the acceptable range of model estimates, Sylvera maintains confidence in the gross removal quantification.

Carbon Score continued

GHG Emission Quantification

To evaluate net carbon removals, Sylvera has developed an in-house Life Cycle Assessment (LCA) model, focusing on key emission sources including feedstock sourcing, energy consumption, and construction. In the updated framework (February 2026), the LCA impact on the carbon score is now calculated proportionally to the entire removal, rather than as a direct deduction. This means that individual emission variations have a smaller impact on the overall carbon score, providing a more representative assessment of a project's total GHG sources, sinks and reservoirs. Sylvera's LCA model has been peer-reviewed by UCL Consultants, and uses the most up-to-date emission factors available from ecoinvent, UK Government and academic literature.

Emission calculations are performed by taking activity data, matching this to an emission factor, and multiplying these together to give an output in CO₂ equivalent (CO₂e):

$$\text{Activity data} * \text{Emission Factor} = \text{kgCO}_2\text{e}$$

	Electricity	1000 kWh		0.27522		275.22 kg CO₂e
	example:	of purchased electricity	×	kgCO ₂ e/kWh	=	

Therefore, by performing these calculations across important emissions sources, Sylvera can assess whether the project is likely to be accurately reporting its GHG emissions.

Feedstock Counterfactual Quantification

Sylvera assesses the counterfactual scenario for feedstock use, determining whether the feedstock would have resulted in removals under a baseline scenario. By examining the type and alternative fate of feedstocks, Sylvera assesses whether the project has appropriately set their baseline value. In cases of uncertainty, conservative deductions are applied to ensure integrity in removal estimates.

Leakage Quantification

Sylvera evaluates the risk of leakage by analyzing the sourcing and usage of feedstocks, as well as potential indirect impacts such as land use changes, activity-shifting or replacement emissions. The assessment ensures that leakage risk is investigated for removal claims. Where data is insufficient, Sylvera applies conservative deductions to uphold rigorous and transparent carbon accounting.

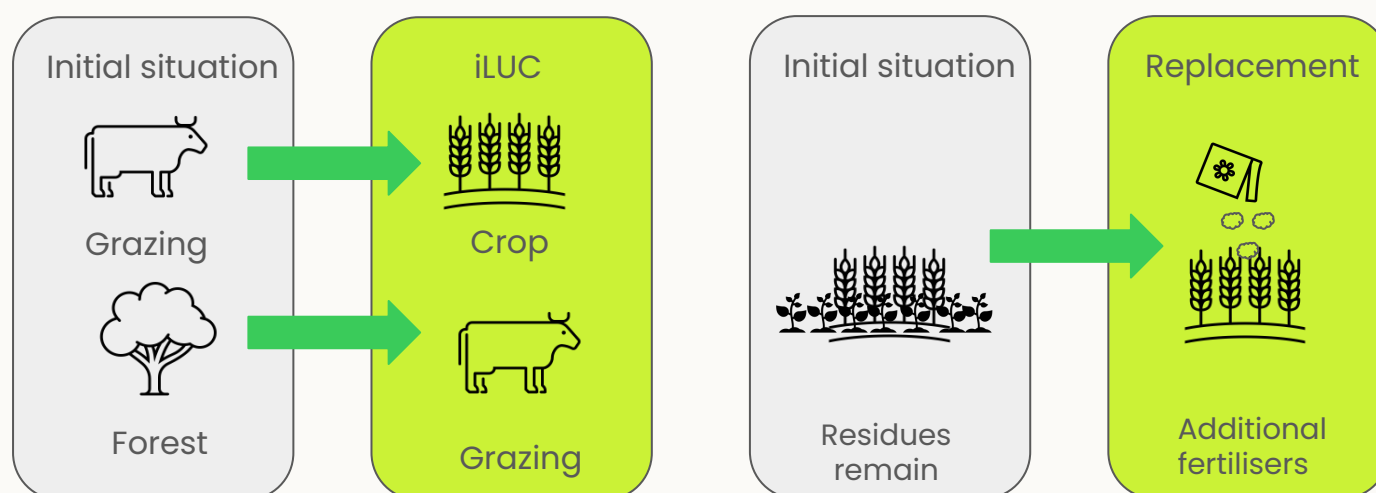


Figure 1. Examples of indirect Land Use Change (iLUC) and Replacement Emissions

Additionality Score

What is it?

Sylvera's additionality score assesses whether (1) a project would or wouldn't have been implemented in the absence of revenue generated through the sale of carbon credits, (2) the project is not mandatory/strongly supported by policies and regulations and (3) the project is not presenting double accounting risk with other VCM activities.

Why does it matter?

Additionality underpins the validity of credits issued by a project. If the project is not additional, then one credit purchased does not equate to one metric ton of carbon removed and therefore yields no climate benefit above the business-as-usual (BAU) scenario.

A project would score high in additionality if Sylvera's financial analysis demonstrates the need for carbon finance to make the project economic. This is not a binary test and the degree of additionality depends on the carbon price required to make it economic. On the other hand, a project would score low in additionality if the revenues from biochar and any possible co-products are enough alone to make the project economically viable.

How does Sylvera assess additionality?

Financial Additionality

The assessment of financial additionality involves determining whether the economic viability of the project relies on carbon revenues. Sylvera examines if the carbon credits revenue bridges the economic viability gap of the project through sensitivity analysis. Biochar projects that generate additional revenue streams by selling biochar or co-products to third parties, Sylvera assesses whether the project would still be economically viable without carbon revenues. If a project is economically feasible without carbon credits, the credits are considered non-additional. This ensures that carbon finance is allocated only to projects that require it to proceed.

Biochar projects can have more than one revenue stream. The biochar itself can be sold at prices based on the quality of the biochar and other transaction parameters. Furthermore, pyrolysis also creates thermal energy that can be sold along with any bio-oils produced during the process. These products can create significant revenues for the project crossing the hurdle rate required to make the project economic. If these revenues are not enough, carbon finance might be required to bridge this gap. However, reliance on high carbon credit prices can make the likelihood of achieving an IRR above the hurdle rate more difficult and unlikely.

A high-quality project demonstrates that carbon finance is necessary for the project to exceed its hurdle rate, while requiring only a modest carbon price to do so. The project should also account for all emissions associated with producing biochar and its co-products, including bio-oil and energy, and allocate them properly across outputs.

Policy & Regulatory Landscape

Sylvera considers the policy and regulatory context to assess additionality. Projects are evaluated to confirm they are not mandated by law or regulation and to identify whether subsidies or government funding diminish their additionality. By integrating any provided subsidies or financial incentives into our economic analysis, Sylvera ensures that carbon credits are issued only where they play a critical role in making the project viable. This safeguards the integrity of credits in regions with supportive policy landscapes.

Double Accounting

To address the risk of double accounting, research are performed to ensure that carbon credits are not claimed by multiple stakeholders. In biochar projects, this involves verifying that the producer and end user of the biochar do not both claim the same carbon benefit. Clear documentation of credit allocation is required to establish ownership and avoid duplication. This approach safeguards the integrity and credibility of the issued credits.

Permanence Score

What is it?

Permanence refers to the risk that the sequestered carbon will later be reversed and released back into the atmosphere. The input of climatic variables, record of past events, project specific conditions and mitigative activities are used to inform the permanence risk scoring. Permanence is conceptualized as a scale that distinguishes the relative degree of non-permanence (or reversal) risk between projects.

For soil application, the permanence of carbon stored in the biochar depends primarily on the H/C ratio and ambient soil temperature, with production temperature acting as an indirect influence through its effect on biochar quality. Soil type, location, and climatic conditions further shape durability: biochar decays faster in tropical climates due to higher soil temperatures, while clay soils can enhance permanence through physical encapsulation within soil aggregates, though high soil moisture may offset this by increasing microbial activity. Projects making claims of 1,000 years or longer via random reflectance are treated with additional scrutiny, as this method remains scientifically contested and unstandardized ([Sanei et al., 2025](#); [Drobniak et al., 2026](#)).

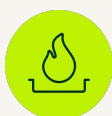
For non-soil applications, the permanence is assessed based on the expected lifespan and end-of-life traceability. This involves examining how these different applications influence the durability and effectiveness of biochar in retaining carbon. Use cases such as construction materials offer strong physical protection during their service life, though end-of-life disposal introduces reversal risk. Applications such as water filtration carry higher near-term reversal risk, as biochar may degrade during active use.

Biochar credits have a low permanence risk given the overall high stability of biochar. The only case where biochar stability is uncertain is where there is a high likelihood and high severity of fire events, however in comparison to nature-based projects, there is significantly less risk of reversal. As a result, Sylvera places greater emphasis on anthropogenic risk, focusing on whether the end-use location is traceable and whether there is clear evidence that biochar has been applied.

How do we calculate the permanence score?

The permanence score is derived from a comprehensive analysis of observational and modeled data, adhering to recognized methodologies and the latest scientific advancements. The permanence of biochar is significantly influenced by its end-use, whether applied in soil or non-soil environments. Sylvera's assessment evaluates biochar stability and decay rate in its specific application while also identifying anthropogenic risks associated with its use.

Stability risk



Random Reflectance

measures the amount of light reflected from biochar, indicating its level of carbonization



Pyrolysis temperature

reflects the temperature at which biochar is produced.



Hydrogen to Carbon (H/C_{org}) ratio

measure of elemental composition, where a lower ratio indicates more stable, aromatic carbon.

Decay rate



Natural risks

Events such as fire, flood, storm can have a direct negative impact on the decay rate of



Soil type and temperature

Soil with high mineral and water content may have higher permanence as it can mineralise and store carbon. In contrast, higher soil temperature leads to faster decay.



Application method

Different application methods such as subsurface soil application, blending with manure or fertilizer etc. will contribute to different levels of permanence.

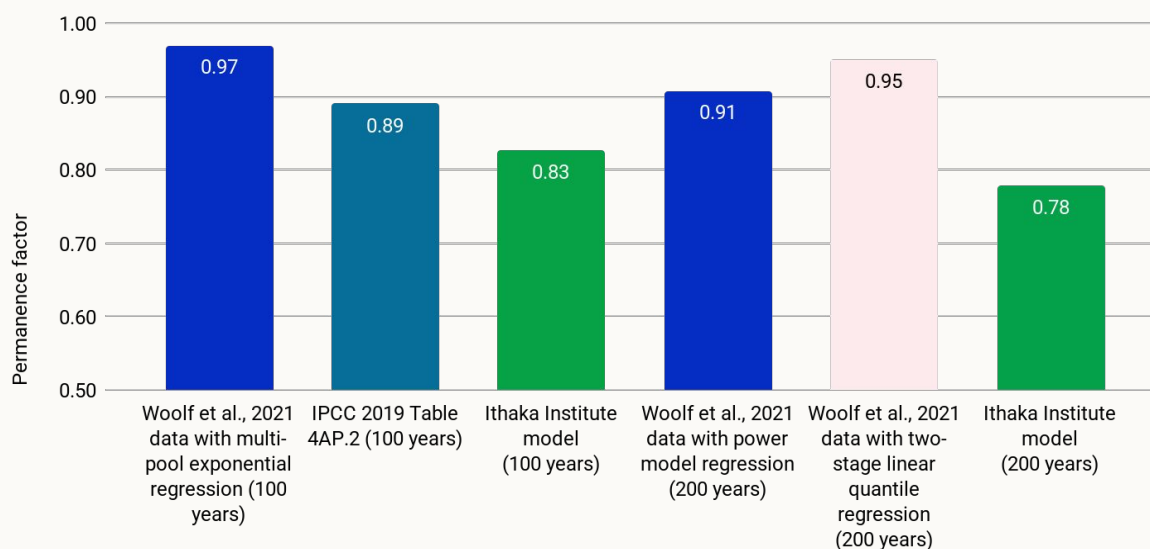
Permanence Score continued

What data do we use in the Permanence Score?

A component of the Stability Rate: the decay rate assessment

Sylvera evaluates permanence by reviewing decay models across peer-reviewed literature and carbon registries to define the full range of plausible permanence outcomes. H/Corg ratio and independently sourced soil temperature data serve as the primary parameters. Project results are benchmarked against scientific and conservative model ranges to assess robustness. Where reported permanence diverges significantly from plausible estimates under equivalent conditions, Sylvera flags an over-crediting risk.

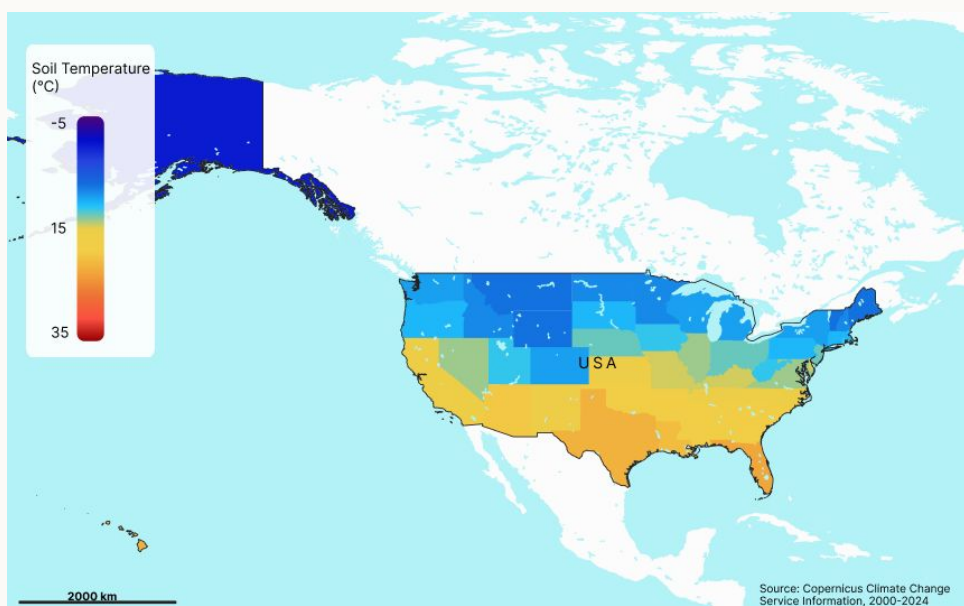
Figure 2. Fraction of biochar carbon remaining after 200 years for a UK project, across reviewed decay models



A component of the Decay Rate: the soil temperature

We leverage a third party data from [Copernicus Climate Change Service \(C3S\) \(2022\)](#) for our soil temperature assessment as seen in figure 3 below. This is to ensure that the soil temperature of biochar's application location claimed by the project is in line with latest verified and established soil temperature data.

Figure 3. Example of soil temperature map for United States of America



Safeguarding & Co-Benefits Score

What is it?

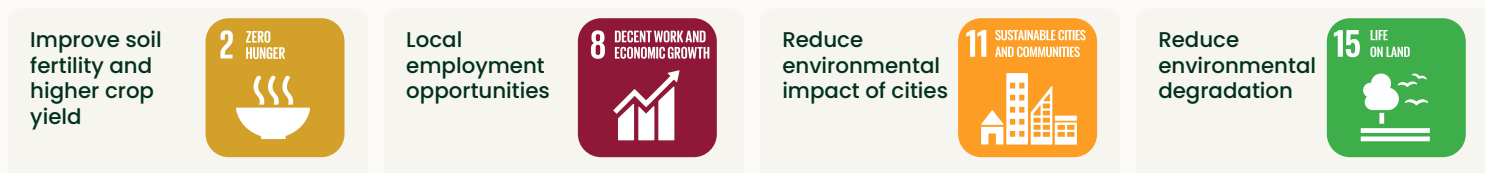
Safeguarding & Co-Benefits Score evaluates whether appropriate community and environmental safeguards are in place and the extent to which the project delivers benefits to local communities and biodiversity beyond its carbon outcomes.

How do we assess the co-benefits of biochar credits?

Sylvera measures the impact biochar project activities have on biodiversity. We leverage data provided by project developers, IUCN data, and IBAT data.

When assessing community impact, we utilize data disclosed by project developers and the Sustainable Development Goals (SDG) to triangulate a project's community impact.

An example of a biochar projects contribution to biodiversity and community:



COMMUNITIES

Social Safeguards

To ensure the project aligns with social safeguard principles, we conduct a thorough evaluation of potential risks, including inadequate worker safety, insufficient community engagement, or inequitable benefit-sharing. As part of our assessment, we carefully analyze stakeholder engagement processes and worker safety practices to measure their impact on local communities. These evaluations are integral to understanding and mitigating social risks while fostering positive outcomes for all stakeholders involved.

Sustainable Development Goals

We independently identify which UN SDGs the project is contributing towards by assessing the activities implemented by the project. We determine the relative impact of activities on local communities by scaling the SDG impact against country-level performance, the size of the population affected, and the carbon removals achieved by the project.

BIODIVERSITY

Threats & Safeguards

We assess whether the facility is located in a conservation area. There are no threats associated such as pollutant treatment with the application of biochar in soil or non-soil applications.

Biodiversity Protection & Feedstock Integrity

We assess the extent to which the project has contributed to biodiversity loss, considering both the broader impacts of the biochar project, feedstock sustainability and traceability, and contributions to biodiversity-related SDGs.

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Sylvera is a carbon data platform on a mission to incentivize investment in real climate action. Global corporations, financial institutions, and governments rely on Sylvera to develop and execute their carbon credit strategies, drive measurable progress toward net zero goals, and optimize returns on investment. Co-founded in 2020 by Dr. Allister Furey and Sam Gill, the company is headquartered in London with additional offices in Belgrade, New York, and Singapore.

To learn more about Sylvera, [contact us](#).

